

Title: Understanding Information-Seeking Behaviour of Singapore Secondary School Students through Google Search Queries: Specific-Broad and Surface-Deep Patterns

Abstract

- **Background/Context:** Define the topic and the specific problem or research gap you are addressing.
- **Purpose/Aim:** State the main objective or research question/hypothesis.
- **Methods:** Briefly describe the research design, participants, and techniques used.
- **Results/Findings:** Summarise the most important findings or trends, including crucial data.
- **Conclusion/Implications:** Explain the significance of the findings, the main takeaway, and potential applications.

With the nationwide rollout of Personal Learning Devices (PLDs), web search has become routine for Singapore secondary school students' schoolwork and self-directed learning. However, there is limited research on analysing query logs to understand how students formulate information needs and how these needs evolve. This study examines students' Google search queries on PLDs and characterises information-seeking behaviour along two dimensions: Specific-Broad (S-B), capturing granularity, and Surface-Deep (S-D), capturing cognitive depth. Queries were extracted from Google URLs, navigational queries were filtered, and query streams were segmented into episodes using a 5-minute inactivity threshold. From 40,586 users, we retained only "active" users with more than 400 episodes to support reliable behavioural modelling and keep computation feasible for embedding- and NER-based features, yielding 4,961 users and 3.38 million queries. For S-B, we engineered episode- and user-level measures spanning lexical and semantic diversity, named-entity diversity, WordNet hypernym specificity, concreteness, and topical breadth via a pretrained 99-topic model (episode topic entropy, topic counts). For S-D, we captured intent cues (closure-seeking, comparative), episode engagement, and within-episode reformulation using normalised Levenshtein distance. Results show that most episodes are single-topic, with a smaller tail of multi-topic sessions. Clustering on user-level specificity features reveals four specificity profiles. Along S-D, closure-seeking (6.4%) and comparative (3%) queries are rare, while reformulation is pervasive and substantial in deeper episodes, suggesting extended sessions reflect iterative refinement rather than repeated identical queries. These findings provide an interpretable, scalable framework for profiling students' information-seeking behaviour and guiding educational interventions that support question-asking and self-directed learning.

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